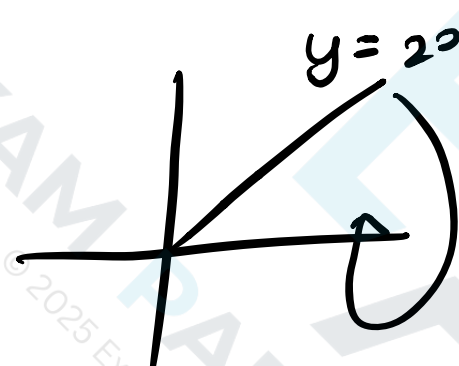


Surface area of revolution

Basic skills

Q1. $V = \int_a^b \pi y^2 dx$

Q2. a)  cone



b)
$$V = \int_0^3 \pi (2x)^2 dx = \int_0^3 4\pi x^2 dx$$
$$= \left[\frac{4}{3} \pi x^3 \right]_0^3 = \frac{4}{3} \pi 3^3 - 0 = 36\pi$$

c) For surface need $\frac{dy}{dx} = 2$ so

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + 2^2} = \sqrt{5}$$

so
$$S = \int_0^3 2\pi \times 2x \times \sqrt{5} dx = \int_0^3 4\sqrt{5}\pi x dx$$
$$= \left[2\sqrt{5}\pi x^2 \right]_0^3 = 2\sqrt{5}\pi 3^2 - 0 = 18\sqrt{5}\pi$$

note. $18\sqrt{5}\pi$ is the curved area



and not the base
which is $\pi \times 6^2 = 36\pi$

d) $r=6, h=3$ for cone

2×3 as $x=3$ at base

$$\begin{array}{c} l \\ \triangle \\ r \end{array} \quad h \quad \begin{aligned} l &= \sqrt{r^2 + h^2} \\ &= \sqrt{36 + 9} \\ &= 3\sqrt{5} \end{aligned}$$

$$\text{so } v = \frac{\pi \times 6^2 \times 3}{3} = 36\pi \text{ as before}$$

$$S = \pi \times 6(6 + 3\sqrt{5}) = \underbrace{36\pi}_{\text{base}} + \underbrace{18\sqrt{5}\pi}_{\text{curved}}$$

Competency skills

Q3. It is not as revolving makes
Circular motions

$$\text{Q4. } y^2 = 4ax \quad x \in [0, a]$$

$$\text{so } y = 2\sqrt{a}\sqrt{x} \Rightarrow \frac{dy}{dx} = \frac{\sqrt{a}}{\sqrt{x}}$$

$$\text{Then } \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{a}{x}} = \frac{1}{\sqrt{x}} \sqrt{x+a}$$

$$\text{Then } S = \int_0^a 2\pi \times 2\sqrt{a}\sqrt{x} \times \frac{1}{\sqrt{x}} \sqrt{x+a} \, dx$$

$$= 4\pi\sqrt{a} \int_0^a \sqrt{x+a} \, dx$$

$$= 4\pi\sqrt{a} \int_a^{2a} \sqrt{u} \, du$$

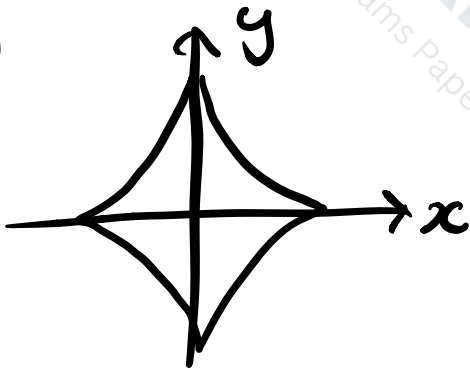
let $u = x+a$
 $du = dx$

$$= 4\pi\sqrt{a} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_a^{2a} = 4\pi\sqrt{a} \left(\frac{2}{3} \sqrt{2a}^3 - \frac{2}{3} \sqrt{a}^3 \right)$$

$$= \frac{8}{3}\pi \left(\sqrt{2}^3 \sqrt{a}^4 - \sqrt{a}^4 \right) = \frac{8}{3}\pi a^2 (\sqrt{2}^3 - 1)$$

Q5. $x(\theta) = a \cos^3 \theta$, $y(\theta) = a \sin^3 \theta$ $0 \leq \theta \leq 2\pi$

a)



Symmetries $x=0, y=0$

b) for S let's use only upper part and then we can just double the positive, positive quadrant.

so only $0 < \theta \leq \frac{\pi}{2}$ $\frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta$

$\frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$ so for $\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}$

$$= \sqrt{9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \sin^4 \theta \cos^2 \theta}$$

$$= 3a \cos \theta \sin \theta \sqrt{\cos^2 \theta + \sin^2 \theta}$$

$$= 3a \cos \theta \sin \theta$$

So

$$S = 2 \int_0^{\frac{\pi}{2}} 2\pi \times a \sin^3 \theta \times 3a \cos \theta \sin \theta d\theta$$

double
quadrant
↺ ↻

$$= 12\pi a^2 \int_0^{\frac{\pi}{2}} \sin^4 \theta \cos \theta d\theta$$

let $u = \sin \theta$
 $du = \cos \theta d\theta$

$$= 12\pi a^2 \int_0^1 u^4 du = 12\pi a^2 \left[\frac{1}{5} u^5 \right]_0^1$$

$$= 12\pi a^2 \left(\frac{1}{5} - 0 \right) = \frac{12}{5} \pi a^2$$

Q6. $x = at^2$, $y = 2at$ $[a, 4a]$

so $a = at^2$ $t = 1$, $4a = at^2$, $t = 2$

so from $t = 1$ to 2

$$\frac{dx}{dt} = 2at \quad \frac{dy}{dt} = 2a \quad \text{so} \quad \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{4a^2 t^2 + 4a^2}$$

$$= 2a\sqrt{t^2 + 1}$$

Then

$$S = \int_1^2 2\pi \times 2at \times 2a\sqrt{t^2+1} dt$$

$$= 8a^2\pi \int_1^2 t\sqrt{t^2+1} dt$$

↓ let $u = t^2 + 1$

$$= 8a^2\pi \int_2^5 \sqrt{u} \times \frac{1}{2} du \quad \text{so } du = 2t dt$$

$$= 8a^2\pi \left[\frac{2}{3} \sqrt{u}^3 \right]_2^5 = 8a^2\pi \left(\frac{2}{3} \sqrt{5}^3 - \sqrt{2}^3 \right)$$

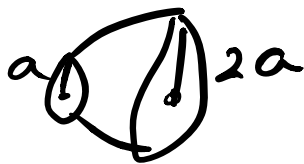
$$= \frac{16}{3} a^2\pi (5\sqrt{5} - 2\sqrt{2})$$

Now need top + bottom to add to curve.

$$\text{At } t=1 \quad y=2a \quad \text{so } \pi \times (2a)^2 = 4a^2\pi$$

$$t=2 \quad y=4a \quad \text{so } \pi \times (4a)^2 = 16a^2\pi$$

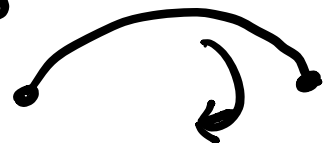
$$\text{so total is } a^2\pi \left(\frac{16}{3} (5\sqrt{5} - 2\sqrt{2}) + 20 \right)$$



Q7. $x = a(t - \sin t)$, $y = a(1 - \cos t)$

One period is $t \in [0, 2\pi]$ so

$$\frac{dx}{dt} = a(1 - \cos t) \quad \frac{dy}{dt} = a \sin t$$



$$\begin{aligned}
 \text{SO } \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} &= \sqrt{a^2(1-\cos t)^2 + a^2 \sin^2 t} \\
 &= a \sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t} = a \sqrt{2 - 2\cos t} \\
 &= \sqrt{2} a \sqrt{1 - \cos t}
 \end{aligned}$$

$$\begin{aligned}
 S &= \int_0^{2\pi} 2\pi \times a(1-\cos t) \times \sqrt{2} a \sqrt{1-\cos t} dt \\
 &= 2\sqrt{2} a^2 \pi \int_0^{2\pi} (1-\cos t)^{\frac{3}{2}} dt
 \end{aligned}$$

$$1 - \cos t = 2 \sin^2\left(\frac{t}{2}\right) \quad \text{as } \cos 2\theta = 1 - 2\sin^2\theta \text{ with } \theta = \frac{t}{2}$$

$$= 2\sqrt{2} a^2 \pi \int_0^{2\pi} \left(2 \sin^2\left(\frac{t}{2}\right)\right)^{\frac{3}{2}} dt$$

$$= 2\sqrt{2} a^2 \pi \int_0^{2\pi} \sqrt{2}^3 \sin^3\left(\frac{t}{2}\right) dt$$

$$= 8a^2 \pi \int_0^{2\pi} \sin^3\left(\frac{t}{2}\right) dt = 16a^2 \pi \int_0^{\pi} \sin^3 u du$$

$\xrightarrow{u = \frac{t}{2}} \quad du = \frac{1}{2} dt$

Now for $\int \sin^3 u du$ use by parts

EXAM PAPERS PRACTICE

$$\text{so } \int \sin^3 u \, du = -\cos u \sin^2 u + \int 2 \cos^2 u \sin u \, du$$

↘ $n = \cos u$

$$= -\cos u \sin^2 u - \int 2n^2 \, dn$$

$$= -\cos u \sin^2 u - \frac{2}{3} n^3$$

$$= -\cos u \sin^2 u - \frac{2}{3} \cos^3 u$$

$m = \sin^2 u$ $v' = -\cos u$
 $m' = 2 \cos u \sin u$ $v = \sin u$

$$\text{so } S = 16a^2 \pi \left[-\cos u \sin^2 u - \frac{2}{3} \cos^3 u \right]_0^\pi$$

$$= 16a^2 \pi \left(0 + \frac{2}{3} - \left(0 - \frac{2}{3} \right) \right) = 16a^2 \pi \times \frac{4}{3}$$

$$= \frac{64}{3} a^2 \pi$$

Advanced skills

Q8.

a) $y = \frac{1}{x}$ so $v = \int_1^\infty \pi x \left(\frac{1}{x} \right)^2 dx$

$$\Rightarrow v = \pi \lim_{k \rightarrow \infty} \int_1^k x^{-2} dx = \pi \lim_{k \rightarrow \infty} \left[-x^{-1} \right]_1^k$$

$$= \pi \lim_{k \rightarrow \infty} \left(-\frac{1}{k} - -\frac{1}{1} \right) = \pi \times (0 + 1) = \pi$$

So finite.

$$b) \frac{dy}{dx} = -\frac{1}{x^2} \text{ so } \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{1}{x^4}} = \sqrt{1 + x^{-4}}$$

$$\text{Then } S = \int_1^{\infty} 2\pi x \cdot \frac{1}{x} \times \sqrt{1 + x^{-4}} \, dx$$

$$= 2\pi \lim_{k \rightarrow \infty} \int_1^k \frac{\sqrt{1 + x^{-4}}}{x} \, dx$$

$$d) \text{ using } \sqrt{1 + x^{-4}} = 1 + \frac{1}{2} x^{-4} + \dots$$

$$S = 2\pi \lim_{k \rightarrow \infty} \int_1^k \frac{1}{x} + \frac{1}{2} x^{-5} + \dots \, dx$$

$$= 2\pi \lim_{k \rightarrow \infty} \left[\ln x - \frac{1}{8} x^{-4} - \dots \right]_1^k$$

$$= 2\pi \lim_{k \rightarrow \infty} \left(\ln k - \frac{1}{8} k^{-4} - \dots - (0 - \frac{1}{8} + \dots) \right)$$

$$= \infty \quad \text{Surface area is not finite as } \ln k \rightarrow \infty$$

$$d) \text{ If } y = \frac{1}{x^2} \quad \frac{dy}{dx} = -\frac{2}{x^3} \text{ so } \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= \sqrt{1 + \frac{4}{x^6}} = \sqrt{1 + 4x^{-6}}$$

Then $S = 2\pi \lim_{k \rightarrow \infty} \int_1^k \frac{1}{x^2} \sqrt{1 + 4x^{-6}} dx$

$$= 2\pi \lim_{k \rightarrow \infty} \int_1^k \frac{1}{x^2} (1 + 2x^{-6} + \dots) dx$$

$$= 2\pi \lim_{k \rightarrow \infty} \int_1^k x^{-2} + 2x^{-8} + \dots dx$$

$$= 2\pi \lim_{k \rightarrow \infty} \left[-x^{-1} - \frac{2}{9} x^{-9} - \dots \right]_1^k$$

$$= 2\pi \lim_{k \rightarrow \infty} \left(-k^{-1} - \frac{2}{9} k^{-9} - \dots - \left(-1 - \frac{2}{9} - \dots \right) \right)$$

$$= 2\pi \left(1 + \frac{2}{9} + \dots \right) \leftarrow \text{check sum is finite}$$

if sum is finite then
yes surface area is